

04 AI Oracles and Semantic Compliance Architecture v3.4 EN

1. Module Purpose and System Scale of MOD-AGO-CME-v3.4-Adaptive

- **Module Identifier**: MOD-AGO-CME-v3.4-Adaptive.
- **Execution Stack**: C++17 / Python FastAPI / ONNX Runtime / Open Enclave SDK inside TEE Enclave.
- **PoUCC Standard (Proof of Useful Cognitive Computation)**: Algorithmic standard governing the distribution of network hardware compute resources. Under PoUCC, 75% of global network GPU capacity is allocated to high-performance linguistic and vector processing, NLP-driven normative act analysis, B2B commercial estimate auditing, and anti-fraud cartel de-anonymization. The remaining 25% of compute capacity is distributed between maintaining DID infrastructure (20%) and supporting the MPC connectivity protocol (5%).

Component	Tech Stack	Function in CME Contour
API & Service Layer	Python FastAPI / ONNX Runtime	Asynchronous vector parsing and local neural model execution
Compute Allocation	PoUCC (75% GPU Clusters)	Semantic decompilation, vector estimate audits, anti-fraud de-anonymization

2. Three-Layer Deconstruction and Admission Threshold Filter

Prior to being submitted for public voting in the Decentralized Agora, every civil or institutional initiative undergoes mandatory automated pre-moderation through the

decentralized network of Semantic AI Oracles. Pre-moderation executes a three-layer vector decompilation of initiative texts and project cost documentation:

- **Layer 1: Jurisprudence ($Comp_{law}$)**: Conducts automated legal audits to evaluate compliance with the Fundamental Human Rights Contour, existing legislative frameworks, and Whitepaper system invariants. The layer detects hidden legal trapdoors, jurisprudential contradictions, discriminatory clauses, and corruption loopholes.
- **Layer 2: Budget Estimates ($Comp_{fin}$)**: Executes vector parsing of financial metrics, estimate specifications, and budgetary requests using cosine similarity methods against reference trust matrices. This layer flags populist cost inflations, unbudgeted expenditure requests, and signs of cartel price-fixing.
- **Layer 3: Ecology ($Comp_{eco}$)**: Performs direct mathematical modeling of the project's comprehensive ecological and thermodynamic footprint. Carbon emissions, acoustic noise, hydrological impact, and thermal pollution vectors are computed using live IoT telemetry and hydro-oracle data feeds.

The **Heaviside Step Admission Threshold Filter** (Ψ) translates linguistic parameters into a strict mathematical decision field by calculating the product of unit step functions across all three layers:

$$\Psi = \text{Step}(Comp_{law} - Th_{law}) \cdot \text{Step}(Comp_{fin} - Th_{fin}) \cdot \text{Step}(Comp_{eco} - Th_{eco})$$

where Th_{law}, Th_{fin}, and Th_{eco} represent established compliance thresholds ($Th_{law} = 0.70$, $Th_{fin} = 0.85$, $Th_{eco} = 0.65$).

- If $\Psi = 1.0$, the proposal satisfies all legal, financial, and ecological criteria and is automatically admitted to the public voting pool.
- If $\Psi = 0$, the proposal fails pre-moderation due to unfulfilled metrics or populist deficit and is routed to the Cognitive Mediation Protocol.

3. Cognitive AI-Architect

The Cognitive AI-Architect module (AI-Architect Engine / Constructive Transformer) eliminates technocratic barriers and citizen alienation from governance, converting the AI Oracle from a rigid censor into an active project design institute. When a civil proposal fails compliance thresholds ($\Psi = 0$) due to missing cost estimates or improper legal formatting, the system does not reject the submission, but automatically triggers the auto-completion procedure:

- **Value Core Extraction**: Cleanses raw submission text of emotional, manipulative, and populist syntax using an NLP Style Sanitizer.
- **Automated 3-Layer ScenarioBrief Construction**:

    - *Layer 1 (Jurisprudence)*: Aligns the initiative with Whitepaper system invariants, legal standards, and human rights constraints.
    - *Layer 2 (Estimates)*: Calculates the complete project budget in T_{net} tokens using benchmark Real-World Asset (RWA) databases and market indicators.
    - *Layer 3 (Ecology)*: Models predicted thermodynamic, carbon, and hydrological footprints using IoT telemetry.
- **Co-Authorship & One-Click Approval**: Delivers the compiled ScenarioBrief package back to the author's Personal Companion (Satellite Bridge) for one-click approval, re-submitting the project to the Agora with a verified compliance rating of $\Psi = 1.0$.

4. Contour Boundary Enforcer and Atomic Proposal Split

The Contour Boundary Enforcer module enforces hardware-level isolation between decision-making logic and voting rights across two strictly separated domains:

- **Value Contour (Ethics, Fundamental Rights, Culture, Humanitarian Manifestos)**:
    - Operates under the Cleisthenic Absolute Isonomy regime, guaranteeing direct equality on the principle of "1 person = 1 vote".
    - Effective voting weight is strictly fixed at $W_{\text{effective}} = 1.0$ for all verified human residents.
    - Meritocratic surcharges (W_{user}), reputation scores ($TRXS$), and financial capital are forcibly zeroed out.
    - Quadratic Voting (QV) and token expenditures (T_{net}) are hardware-blocked at the kernel level.
- **Mathematical / Economic Contour (Schedules, Budget Estimates, RWA Infrastructure, IT)**:
    - Operates under Socratic Dynamic Meritocracy, where voting weight is bounded within $W_{\text{user}} \in [0.01; 5.00]$ based on Proof-of-Contribution.
    - Capital dominance and commercial lobbying are suppressed through Quadratic Voting, where vote purchasing costs scale quadratically: $C(V) = V^2$. Total expenditure for N votes follows $C_{\text{total}}(N) = \frac{N(N+1)(2N+1)}{6}$.

When an author attempts to combine a budget-intensive economic proposal with a humanitarian manifesto, the module triggers an **Atomic Proposal Split**:

1. The humanitarian manifesto is routed to the Value Contour ($W_{\text{effective}} = 1.0$, No QV).
2. The executive feasibility study and financial estimate are routed to the Mathematical Contour ($W_{\text{user}} \in [0.01; 5.00]$, QV, 3-layer ScenarioBrief).

3. Financial release from treasury escrows is unlocked strictly upon the dual approval of both linked acts.

5. Affiliation Detector and Guild Protection

The Semantic AI Oracle continuously executes vector parsing of tender documentation and cost estimates to identify hidden cartel collusions, corporate affiliations, and discriminatory specifications:

- **Affiliation Detection**: Evaluates cosine similarity across proposal vectors, IP addresses, Edge AI telemetry, and ZK-KYC view keys beneficiary overlaps to detect undisclosed affiliations.
- **Guild Protection & Double-Blind Arbitration**:
    - Expert reviews by meritocratic Guilds utilize double-blind rotation and ZK-commitments to prevent reviewer targeting or bribery.
    - The NLP Style Sanitizer strips unique stylistic markers from submission documents to ensure impartial evaluation.
- **Margin Discounting & Financial Penalty**: When undisclosed commercial affiliation or profit margin inflation above normative thresholds is detected, required collateral increases exponentially ($C(V) = V^2$), and excess commercial profit margin is discounted and redirected to the Regeneration Reserve Fund (S_{reserve}).

6. Two-Level Gradient Filter 'Physics vs Failure' and PoUO Protocol

To enforce personal accountability for reform authors without punishing them for unpreventable systemic disasters, the system incorporates the Two-Level Gradient Filter within the Proof-of-Outcome (PoUO) framework:

- **Status PENDING Activation**: If an enacted policy causes a drop in systemic success metrics ($I_{sr} < 1.0$), the smart contract transitions the project into `PENDING` status rather than executing an immediate binary slashing.
- **Environmental Correlation Analysis (K_{env})**: The system polls Edge AI nodes across adjacent spatial zones to calculate the Environmental Correlation Coefficient (K_{env}), separating author negligence from exogenous physical events (e.g., electronic warfare/EW, cyberattacks, natural disasters).
    - *Exogenous Force Majeure*: If K_{env} confirms external physical disruption, a "Protective Freeze" is granted, suspending penalties during a 72-hour grace period for blind guild arbitration.
    - *Author Negligence / Sabotage*: If force majeure is absent, a cascade degradation of the author's electoral weight is triggered down to $W_{\text{user}} = 0.01$, and staked Soulbound assets are slashed into S_{reserve}.

- **Proof-of-Useful-Outcome (PoUO)**: Advanced competence increments (ΔI_{cp}) and project funding tranches are granted via Tranche Staging, where final financial clearing is unlocked strictly upon verifying long-term operational success and zero environmental degradation.